

which removes the need for multiple code bases. Hybrid apps are able to harness most device features. Hence, the recent trend in smartphone app development is towards the cross-platform hybrid category.

To assist developers in increasing the speed of app development, there are various app development frameworks. Ionic is a specialised UI framework for smartphone app development. The UI frameworks for hybrid smartphone apps are increasing. Figure 1 shows the wide and rich spectrum of UI frameworks currently available.

Each of these frameworks has its own speciality, apart from the common features. For example, *Famo.us* focuses on graphics rendering and is more suitable for development of smartphone apps loaded with animation. The Onsen UI allows developers to build apps for mobiles and tablets with multiscreen support.

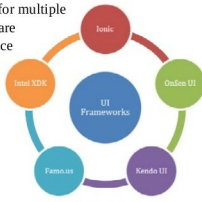


Figure 1: Mobile UI frameworks

An introduction to Ionic

As stated earlier, Ionic is a UI framework. The point to be noted here is that Ionic is not a replacement for frameworks like PhoneGap, etc. Ionic is a front-end SDK, which primarily aims to achieve a native look and feel for the smartphone app's user interface. The Ionic framework is open source with an MIT licence. It fills that vacuum which lies between creating native looking apps and Web standards.

The Ionic apps can be compiled with Cordova or PhoneGap, and stunning native featured apps can be built. These hybrid apps are not dependent on mobile Web browsers but they use the low-level browser shell like WebView on Android, and UIWebView on iOS. The major advantage of using Ionic is that the developers can leverage their familiarity with existing Web technologies like HTML, CSS and JavaScript.

Installation

The Ionic installation requires certain dependencies. You should have *Node.js* installed in your system (<https://nodejs.org/en/>). As Ionic requires the Cordova framework, it needs to be installed, as follows:

```
npm install -g cordova
```

If you are inside a proxy server, the proxy settings must be made using the following commands:

```
npmconfig set proxy http://“user_id:password”@proxy_ip:port
```



Figure 2: Ionic app template types

```
npmconfig set https-proxy http:// “user_id:password”@proxy_ip:port
```

Then the Ionic framework must be installed with the following command:

```
npm install -g ionic
```

In Linux environments, the *sudo* prefix should be used with the above commands. The installation of Ionic is a very simple process which can be completed within minutes, provided you have good Internet connectivity.

The Ionic app-building process

The app-building process in Ionic is very simple. The first step is to start a project and specify an app template type. Ionic has three template types for applications as shown in Figure 2.

To create a blank app, use the following command:

```
$ ionic start todo blank
```

In the above command, 'todo' indicates the app name and the blank indicates the template type. If you are inside the proxy server, then the corresponding proxy settings need to be done before executing the Ionic command. The proxy settings will be done as follows:

```
set proxy=http://“user_id:password”@proxy_ip:port
sethttps_proxy=http://“user_id:password”@proxy_ip:port
```

The successful execution of the above command will create a folder termed 'todo', the structure of which is shown in Figure 3.

The next step is to add the platform support for iOS, Android, etc. This can be carried out with the following commands:

```
$ ionic platform add android
$ ionic platform add ios
```

The next steps are to execute the build and emulate platform-specific commands as shown below:

```
$ ionic build android
$ ionic emulate android
```